

Identifying Risk-Neutral Densities from an Empirical Measure of Listed Quotes

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Abstract

European option prices of a single maturity encode the risk-neutral density of the underlying at expiry. Differentiating those prices twice is to recover the density ill-posed on a finite strike list. The present manuscript rescales the calls to a complementary distribution function and interpolates it with a natural cubic spline. A Gaussian damping of the associated characteristic function inverts in closed form and supplies both the density and a call price. The resulting estimator is in-fill consistent and its repricing accuracy on four listed chains is competitive vis-à-vis existing estimators.

Keywords: risk-neutral density, option prices, kernel estimation, in-fill consistency, state-price density

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1. Introduction

Breeden and Litzenberger (1978) identified the risk-neutral density of the underlying at a fixed expiry with the second strike derivative of the corresponding European call, scaled by a discount factor. The identity assumes a twice differentiable price curve. A market supplies a finite ordered list of strikes and a quote at each knot. Numerical second differentiation of that list is ill-posed: interpolants oscillate, change sign, and depend on the interpolating scheme. The literature responds with parametric mixtures and Gram–Charlier expansions, implied-volatility interpolants, local polynomials of the call, and Bondarenko (2003)’s

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positive convolution approximation, which forces the density to be a mixture of a positive kernel with a discrete probability fitted to prices. Section 2 reviews these approaches.

This paper estimates the density from a single-maturity strip by rescaling the calls to a complementary distribution function, interpolating that function with a cubic spline, and inverting a Gaussian damping of the associated characteristic function in closed form. Completing the unquoted tails and combining two kernel widths improves accuracy further. As the strike mesh fills a compact interval and the bandwidth tends to zero, the estimator converges to the true density on the interior of that interval.

We compare our method to several extant methods. Bondarenko’s method is a sieve of fixed dimension: it does not become finer as more strikes arrive, and it is a density by construction. Yatchew and Härdle (2006) replace the quoted calls by the nearest decreasing convex list; the second differences of that list are a discrete nonnegative measure, not a curve. Aït-Sahalia and Duarte (2003) smooth the same constrained list with a local polynomial, so the density is nonnegative by design and the call cannot match quotes that violate convexity. The estimator of this paper returns a closed-form smooth estimate of $f_{\mathbb{Q}}$ and, from the same smoothing, a call interpolant. On the listed chains of Section 7 the out-of-the-money root-mean-square error (RMSE) sits next to the projection of Yatchew and Härdle (2006) and the mixture of Bondarenko. The cubic of Priestley and Chao (1972) uses the same bandwidth, (18). The interpolant can lie below the Yatchew and Härdle floor because it is not required to be convex.

The remainder of this manuscript is structured as follows. Section 2 reviews related work. Section 3 derives the characteristic function of the normalized call and the closed-form estimator, and compares it to kernel smoothing of the call. Section 4 records two refinements: completing the tails and twicing. Section 5 chooses the bandwidth. Section 6 proves in-fill consistency and illustrates it on exact Heston and variance-gamma quotes. Section 7 presents an empirical illustration that evaluates our estimator against four alternative methods on four listed chains. Section 8 concludes.

2. Related Work

The identity of Section 3 is the continuous-strike version of a spanning argument of Ross (1976): if one could trade a put at every strike, the second derivative of the put curve would be a density, and one could synthesize any sufficiently regular payoff. Dupire (1994) obtained the analogous result in calendar time, which

produces a local-volatility equation. Observed markets supply neither a continuum of strikes nor noiseless quotes.

A first family of methods specifies the density of S_T as a parametric curve and fits the implied option prices to the quotes. Ritchey (1990) and Melick and Thomas (1997) take the density to be a mixture of lognormals, so that the call price is a mixture of Black–Scholes prices; Bahra (1997) and Söderlind and Svensson (1997) use related specifications for interest-rate and foreign-exchange options. Two or three components typically suffice to generate a unimodal, skewed density, but the fit is poorly identified once more than two components are admitted, and the approximation to a Heston or jump-diffusion density need not vanish as more quotes arrive. Expansion methods perturb a baseline density—usually lognormal—by Gram–Charlier or Hermite terms (Jarrow and Rudd, 1982; Corrado and Su, 1996; Rompolis and Tzavalis, 2007; Xiu, 2014). The expansions can match a finite number of moments, but they are not guaranteed to stay nonnegative, and the truncated series can oscillate in the tails.

A second, and in applied work the most common, family interpolates the Black–Scholes implied-volatility function of strike and converts the interpolant back to prices before differentiating twice. Shimko (1993) fitted a quadratic in strike; Malz (1997) used a cubic spline in delta; Bliss and Panigirtzoglou (2002) a smoothing spline. Figlewski (2010) completes the tails with a generalized extreme-value distribution. Because implied volatility is a much flatter function of strike than the call itself, a low-order interpolant often has small bias when quotes are accurate. The drawback is that implied volatility is a nonlinear transformation that inflates price noise where the derivative of price with respect to implied volatility is small, i.e. in the tails. An interpolating spline then treats those noisy tail quotes as exact, and the second derivative of the resulting call curve oscillates and can become negative. Shape-constrained smoothers of the implied-volatility surface (Fengler, 2005, 2009; Gatheral and Jacquier, 2014; Ludwig, 2015) address the sign problem at the cost of a more involved optimization.

A third family estimates the call as a function of strike by kernel regression or local polynomials and obtains the second derivative from the fit. Priestley and Chao (1972) fitted a kernel to observed function values at the design points; the risk-neutral density is obtained by differentiating that fit twice. Aït-Sahalia and Lo (1998) estimated the call as a function of (S, K, T, r, q) by kernel regression; Aït-Sahalia and Lo (2000) used the estimator to recover an implied risk-aversion function. The method is data intensive and, in the original application, pooled option prices across many trading dates. Aït-Sahalia and Duarte (2003) devel-

oped a local polynomial estimator that imposes monotonicity and convexity of C in K , hence nonnegativity of the density. Related constrained least-squares estimators are studied by Yatchew and Härdle (2006) and Birke and Pilz (2009); Grith et al. (2012) survey the kernel literature. Fan and Mancini (2009) combine a parametric guide with a nonparametric correction. In each case the density is the second strike derivative of a smoother of C . The estimator of Section 3 rescales the calls to a complementary cumulative distribution function (cdf), interpolates that function by a cubic spline, and smooths the spline with a Gaussian kernel. The convolution is closed form. Section 3.2 records the comparison. Section 7 implements the cubic of Section 3.2 at the bandwidth (18), and the two-step procedure of Aït-Sahalia and Duarte: a least-squares projection of the quoted calls onto the decreasing convex cone, followed by a local polynomial of the constrained values whose pricing bandwidth is chosen the same way.

Bondarenko (2003) proposed the positive convolution approximation (PCA). The density is required to lie in the set of functions obtained by convolving a discrete probability with a positive kernel, and the discrete probability is chosen to minimize the sum of squared pricing errors. The kernel enforces smoothness and, together with positive weights, guarantees that the result is a density. In Bondarenko’s comparison, the method dominated mixtures of lognormals, Hermite expansions, and several regularization methods. Subsequent simulation evidence on affine jump-diffusions is consistent with this ranking (Lu, 2019). Section 7 implements a log-price analog: forty-one lognormals of common width, centered on a uniform grid in log strike from the first quoted strike to the last, fitted by non-negative least squares. Sum-to-one and forward-mean penalties stay penalties; the weights are not rescaled after the fit. The width is chosen by even/odd pricing error. The same section chooses the Yatchew and Härdle roughness penalty the same way.

Rubinstein (1994) and Jackwerth and Rubinstein (1996) recovered probabilities on a binomial tree by minimizing a distance to a prior subject to pricing constraints. Stutzer (1996) and Buchen and Kelly (1996) maximized entropy. These methods produce a valid density by construction, but the finite-support discretization and the choice of prior affect the tails. Viewing the problem as a linear inverse problem, Cont and Tankov (2004) calibrated jump-diffusion models under a relative-entropy penalty; Monnier (2014) used the singular value decomposition of a restricted put operator.

A number of papers compare methods on market data, where the true density is unknown, so that the criterion is typically pricing error or stability over time (Cooper, 1999; Jondeau and Rockinger, 2000; Coutant et al., 2001; Anagnou

et al., 2005; Bliss and Panigirtzoglou, 2002; Liu et al., 2007). Studies that generate options from a known process are less common. Bondarenko (2003) finds that his method outperforms mixtures, expansions, and several regularization methods on integrated squared error. Lai (2014) finds that kernel regression outperforms spline interpolation and neural nets on pricing error. Santos and Guerra (2015) compare mixtures, Edgeworth expansions, hypergeometric functions, and a Bliss and Panigirtzoglou smoothing spline against Heston and other targets. Lu (2019) compares methods on affine jump-diffusions and again finds Bondarenko's method accurate. The typical ranking is that implied-volatility interpolants price well when quotes are clean, that mixtures are unstable once more than two components are admitted, and that Bondarenko's method is the most accurate among methods that do not use a parametric density of S_T , when the true density is known. Section 6 is in this vein: the targets are Heston and variance-gamma, and the true density is known. Section 7 scores listed index chains, where the criterion is pricing error, against Bondarenko, Yatchew and Härdle, Aït-Sahalia and Duarte, and Priestley and Chao.

3. Construction of the Estimator

3.1. Construction

Assume a risk-neutral measure $\mathbb{Q}$ under which S_T has a density $f_{\mathbb{Q}}$ on $(0, \infty)$ and $\mathbb{E}^{\mathbb{Q}}[S_T] < \infty$. Let r be the continuously compounded interest rate and q the continuous dividend yield. Write $F = S_0 e^{(r-q)T}$ for the forward. Under $\mathbb{Q}$,

$$\begin{aligned} S_0 e^{-qT} &= e^{-rT} \mathbb{E}^{\mathbb{Q}}[S_T] \\ &= e^{-rT} \int_0^{\infty} x f_{\mathbb{Q}}(x) dx, \end{aligned} \tag{1}$$

hence $\mathbb{E}^{\mathbb{Q}}[S_T] = F$. Write C for the corresponding call curve:

$$\begin{aligned} C(K) &= e^{-rT} \mathbb{E}^{\mathbb{Q}}[(S_T - K)^+] \\ &= e^{-rT} \int_K^{\infty} (x - K) f_{\mathbb{Q}}(x) dx. \end{aligned}$$

Now define the normalized option price

$$\tilde{C}(K) \equiv \frac{C(K)}{S_0 e^{-qT}} = \frac{\int_K^{\infty} (x - K) f_{\mathbb{Q}}(x) dx}{F}. \tag{2}$$

The normalized call price is a decreasing function of K : it is equal to 1 if $K = 0$ and equal to 0 if $K = \infty$. Thus the normalized call price, as a function of K , is a complementary cdf, say $\tilde{C}(K) = 1 - P(K)$. The characteristic function (hereafter c.f.) of the law of P is the Fourier–Stieltjes transform

$$\varphi_P(s) = \int_{(0,\infty)} e^{isK} dP(K). \quad (3)$$

For a cdf on $[0, \infty)$ and $s \neq 0$, (3) is

$$\varphi_P(s) = 1 + is \int_0^\infty e^{isK} \tilde{C}(K) dK.$$

Substitute (2). Fubini yields

$$\int_0^\infty e^{isK} \tilde{C}(K) dK = \frac{1}{F} \int_0^\infty f_{\mathbb{Q}}(x) \left[\int_0^x e^{isK} (x - K) dK \right] dx.$$

The inner integral is $-x/(is) + (e^{isx} - 1)/(is)^2$. Hence, for $s \neq 0$,

$$\begin{aligned} \varphi_P(s) &= 1 + \frac{is}{F} \int_0^\infty f_{\mathbb{Q}}(x) \left[-\frac{x}{is} + \frac{e^{isx} - 1}{(is)^2} \right] dx \\ &= 1 - F^{-1} \mathbb{E}^{\mathbb{Q}}[S_T] + \frac{\varphi_{\mathbb{Q}}(s) - 1}{is F} \\ &= \frac{\varphi_{\mathbb{Q}}(s) - 1}{is F}, \end{aligned} \quad (4)$$

where $\varphi_{\mathbb{Q}}(s) = \mathbb{E}^{\mathbb{Q}}[e^{isS_T}]$ and $\mathbb{E}^{\mathbb{Q}}[S_T] = F$ by (1). Solving for $\varphi_{\mathbb{Q}}$, one obtains

$$\varphi_{\mathbb{Q}}(s) = 1 + is F \varphi_P(s). \quad (5)$$

Read from the right, (4) is a pricing formula: a model that supplies $\varphi_{\mathbb{Q}}$ yields φ_P , hence the complementary cdf $\tilde{C} = 1 - P$ and the call (Lewis, 2000). This paper inverts the identity: φ_P is constructed from the quotes, and $\varphi_{\mathbb{Q}}$ is recovered.

The c.f. $\varphi_{\mathbb{Q}}$ is that of S_T . Fourier inversion yields

$$f_{\mathbb{Q}}(K) = \frac{1}{2\pi} \int_{-\infty}^\infty e^{-isK} \varphi_{\mathbb{Q}}(s) ds. \quad (6)$$

Substitute (5) into (6):

$$f_{\mathbb{Q}}(K) = \frac{1}{2\pi} \int_{-\infty}^\infty e^{-isK} ds + \frac{F}{2\pi} \int_{-\infty}^\infty e^{-isK} (is) \varphi_P(s) ds.$$

The first integral is a Dirac mass at $K = 0$. For $K > 0$,

$$f_{\mathbb{Q}}(K) = \frac{F}{2\pi} \int_{-\infty}^{\infty} e^{-isK} (is) \varphi_P(s) ds. \quad (7)$$

Observed calls at strikes $0 < K_1 < \dots < K_m$ supply values $\tilde{C}_i = C_i/(S_0 e^{-qT})$ and hence $P_i = 1 - \tilde{C}_i$. The estimator interpolates these knots by a natural cubic spline P^{cs} . The spline matches value and first derivative at the knots. The c.f. of this absolutely continuous measure is $\int e^{isK} P^{cs'}(K) dK$. Replace φ_P by the product of that integral with the Fourier transform of a Gaussian kernel of width h ,

$$\hat{\varphi}_P(s) = \left(\int_{(0,\infty)} e^{isK} P^{cs'}(K) dK \right) e^{-h^2 s^2/2}. \quad (8)$$

The Gaussian density and its distribution function are

$$\kappa_h(u) = \frac{1}{h\sqrt{2\pi}} \exp\left(-\frac{u^2}{2h^2}\right), \quad \mathcal{K}_h(u) = \int_{-\infty}^u \kappa_h.$$

The factor $e^{-h^2 s^2/2}$ is convolution against κ_h , and multiplication by is is differentiation up to sign. Substituting (8) into (7) therefore yields

$$\hat{f}_{\mathbb{Q}}(K) = -F (P^{cs'} * \kappa_h)'(K), \quad \hat{P}(K) = (P^{cs'} * \mathcal{K}_h)(K). \quad (9)$$

On each cell the integrand is a quadratic times a Gaussian. The integrals of 1, u , and u^2 against ϕ and Φ are elementary, so (9) is closed form. Calls are $\hat{C} = S_0 e^{-qT} (1 - \hat{P})$. These knots are the quoted strikes. Section 4.1 adds the endpoints that complete the distribution.

3.2. Comparison to Naive Kernel Smoothing

In the continuum, Breeden and Litzenberger plus a kernel is unambiguous. If C is the true call curve,

$$(C * \kappa_h)'' = C'' * \kappa_h = e^{-rT} (f_{\mathbb{Q}} * \kappa_h). \quad (10)$$

Smoothing C and differentiating twice is a kernel smooth of $f_{\mathbb{Q}}$. On a strike list the discretization matters.

A Priestley and Chao estimator treats the observations as levels C_i and weights them by the design spacing $\Delta K_i = K_i - K_{i-1}$ ($K_0 := K_1 - (K_2 - K_1)$):

$$\hat{C}^{PC}(K) = \sum_{i=1}^m \Delta K_i C_i \kappa_h(K - K_i), \quad \hat{f}_{\mathbb{Q}}^{PC}(K) = e^{rT} \sum_{i=1}^m \Delta K_i C_i \kappa_h''(K - K_i), \quad (11)$$

where $\kappa_h''(u) = (u^2/h^2 - 1)h^{-2}\kappa_h(u)$. The equivalent kernel for the density is κ'' . The interpolant is a weighted sum of the observed prices.

That sum is a Riemann approximation to (10). It treats the observed levels as atoms of a measure. They are not: C is a smooth function of K , observed at the knots. Convolution requires a function on the line. A linear interpolant of C has second derivative zero almost everywhere and Dirac masses at the kinks, so it cannot recover f_Q . The usual construction is a cubic spline C^{cs} through the quotes, extended by linear wings of slope in $[-e^{-rT}, 0]$:

$$\hat{f}_Q^{\text{cs}}(K) = e^{rT} (C^{\text{cs}} * \kappa_h)''(K). \quad (12)$$

A cubic is the lowest-degree interpolant whose second derivative is not identically zero on each cell; the same degree appears in local-polynomial estimation of a second derivative (Aït-Sahalia and Duarte, 2003). Without the interpolant, (11) places a large weight on κ'' wherever C is large, including deep in the money where C is nearly linear and C'' is nearly zero.

Equation (9) is the same convolution applied to the derivative of a cubic spline of P .

On an equal grid with already monotone C , $\Delta C_i \approx C'(K_i)\delta$, so (11) and (9) are close when the spline of P is nearly linear on each cell. They are not close when C is large and locally linear, as a deep in-the-money call is. Section 7 implements the cubic interpolant (12) on listed chains.

4. Two Refinements

4.1. Completing the Tails

The spline of Section 3.1 stops at the first and last quotes. On $(0, K_1)$ it does not yet specify P , and if $C_m > 0$ the residual mass $1 - P(K_m)$ is unplaced. Both ends are wrong as a cdf on $(0, \infty)$.

On $(0, K_1]$ a deep in-the-money call is intrinsic, $C(K) = e^{-rT}(F - K)$, hence $\tilde{C}(K) = 1 - K/F$ and $P(K) = K/F$. That is a straight line through the origin. The estimator matches the observed $P(K_1)$ rather than K_1/F ,

$$P(K) = P(K_1) \frac{K}{K_1}, \quad 0 < K \leq K_1. \quad (13)$$

When the first quote is at intrinsic, $P(K_1) = K_1/F$ and (13) is exactly K/F . The spline imposes the endpoint $P(0) = 0$, which is the same line at the boundary.

On the right, $\tilde{C}(\infty) = 0$ so $P(\infty) = 1$. If $C_m > 0$, a European call still has to fall to zero. The residual call is continued linearly with slope in $[-e^{-rT}, 0)$ until

$C = 0$ at an intercept K_{end} . The slope is the last strictly negative step of the quoted call, clipped to that interval. A flat last step, two quotes on the premium floor, is not a decay: using it would send K_{end} to the cap $8F$. The spline imposes the single endpoint $P(K_{\text{end}}) = 1$. The bandwidth still counts a notional knot spacing along that line, eight per last quoted gap,

$$n_R = \text{round}(8(K_{\text{end}} - K_m)/(K_m - K_{m-1})), \quad (14)$$

are equally spaced in $(K_m, K_{\text{end}}]$ and converted to $P = 1 - \tilde{C}$. If $C_m = 0$, $P(K_m) = 1$ already and this wing is empty.

Only the left completion changes the estimator on the Heston and variance-gamma chains below: the first quoted call is at intrinsic, and the last quoted call is already zero (dense) or of order 10^{-5} (sparse).

4.2. Twicing

The Gaussian kernel κ_h has second moment h^2 , so the leading bias of (9) is $O(h^2)$. When the quotes are treated as exact there is no observation-error variance to inflate, and a higher-order kernel is admissible. Twicing (Schucany and Sommers, 1977) replaces the kernel transform $e^{-h^2 s^2/2}$ in (8) by

$$2e^{-h^2 s^2/2} - e^{-h^2 s^2}, \quad (15)$$

which is the Fourier transform of $2\kappa_h - \kappa_{h\sqrt{2}}$. The equivalent kernel has vanishing second moment, so the $O(h^2)$ bias becomes $O(h^4)$. In closed form,

$$\hat{f}_{\mathbb{Q}}^{\text{tw}}(K) = 2\hat{f}_{\mathbb{Q},h}(K) - \hat{f}_{\mathbb{Q},h\sqrt{2}}(K), \quad (16)$$

and the same combination is applied to the interpolant in (9). The equivalent kernel takes negative values, so $\hat{f}_{\mathbb{Q}}^{\text{tw}}$ is not a density. The call is $S_0 e^{-qT}(1 - \hat{P}^{\text{tw}})$, with $\hat{P}^{\text{tw}}$ clipped to $[0, 1]$ and both premiums floored at zero.

Twicing a truncated distribution amplifies leakage from the missing tails, so it is applied only after Section 4.1. The estimator used below is (9) with those tails and with (16).

5. Choice of Bandwidth

The width h is the only tuning constant in (16). Two rules are used below.

With exact quotes the bias is the usual $O(h^2)$ kernel bias, or $O(h^4)$ after twicing, and the spline error vanishes with the mesh. For a Gaussian it is enough that h be

a small multiple of the typical quoted gap. Write δ for the median of $\{K_i - K_{i-1}\}$ on the quoted strikes, not on the filled wings, and take

$$h = \min(1.2 \delta, 0.30 (K_m - K_1)). \quad (17)$$

The cap binds only on a very short chain. On an equally spaced grid every gap equals δ . This is the rule in Section 6.1, where the quotes are exact.

When quotes are noisy, (17) inherits the noise in the jumps. A listed chain is a noisy sample of the call curve, not an in-fill of an exact function. The object being smoothed is then closer to a density of the jumps than to a second derivative from a vanishing mesh. Replacing the mesh by the width of S_T and using the Gaussian exponent of Silverman (1986), $n^{-1/5}$, yields

$$h = c \times 1.06 F \hat{\sigma}_{\text{ATM}} \sqrt{T} n^{-1/5}, \quad (18)$$

where $\hat{\sigma}_{\text{ATM}}$ is the at-the-money (ATM) Black–Scholes implied volatility, the median of contracts with $|K - F| \leq 0.03F$ (or the median of all finite implied volatilities if that window is empty), and n is the number of masses after the right wing of Section 4.1, not the left filler and not necessarily the number of listed quotes. The scale is the scale of S_T , not the standard deviation of the jumps Δ_j . The ordinary Silverman rule computed from those jumps oversmooths, because that standard deviation is of the order of the chain width. The empirical illustration below uses (18) with $c = 0.55$ for both our estimator and Priestley and Chao. Theorem 1 below requires the mesh and the bandwidth both to tend to zero. On a sequence of grids that fill a fixed interval with mesh $\delta \propto n^{-1}$, both (17) and (18) satisfy this requirement.

6. In-Fill Consistency

Write P for the complementary-call cdf of Section 3, so that $f_{\mathbb{Q}} = -F P''$ on $(0, \infty)$. The estimator (9) replaces P' by the derivative of its cubic spline and smooths with κ_h .

Theorem 1. *Let $\mathbb{E}^{\mathbb{Q}}[S_T] < \infty$, and let $f_{\mathbb{Q}}$ be four times continuously differentiable on $(0, \infty)$. Let C be the corresponding call curve. Fix a compact interval $[A, B] \subset (0, \infty)$. Consider a sequence of exact quote grids $0 < K_1^{(m)} < \dots < K_m^{(m)}$ with mesh $\delta_m = \max_i (K_i^{(m)} - K_{i-1}^{(m)}) \rightarrow 0$, $K_1^{(m)} \leq A$ and $K_m^{(m)} \geq B$. Form the cubic spline of Section 3.1 and complete the tails as in Section 4.1. Let $h_m \rightarrow 0$. Then both (9) and its twiced version (16) converge to $f_{\mathbb{Q}}$ uniformly on every compact subset of (A, B) .*

Proof. The estimator is $\hat{f}_{\mathbb{Q}} = -F(P^{cs'} * \kappa_h)'$, and $f_{\mathbb{Q}} = -FP''$. Split

$$(P^{cs'} * \kappa_h)' - P'' = ((P^{cs'} - P') * \kappa_h)' + (P' * \kappa_h - P')'.$$

The second term is ordinary kernel bias of $P'' = -f_{\mathbb{Q}}/F$. Because κ is even, $\int \kappa = 1$ and $\int u\kappa(u)du = 0$. If $f''_{\mathbb{Q}}$ is bounded on a neighborhood of $[A, B]$, the difference is $O(h^2)$ uniformly on compact subsets of (A, B) several bandwidths away from the endpoints. The twiced kernel $2\kappa_h - \kappa_{h\sqrt{2}}$ has Fourier transform (15), hence vanishing second moment, so the same expansion starts at order h^4 whenever $f_{\mathbb{Q}}$ is C^4 .

The first term is interpolation. On a quasi-uniform mesh, the natural cubic spline of a C^4 function converges to the function, and its first derivative converges to the first derivative, uniformly on compact subsets of the interior of the grid. Thus $P^{cs'} \rightarrow P'$ as $\delta_m \rightarrow 0$. Convolution against κ_h preserves the limit, and one derivative in K does not change the conclusion on those compacts.

The filled wings are supported on $(0, K_1]$ and, if $C_m > 0$, on $(K_m, K_{\text{end}}]$. Their contribution at an evaluation point $K \in [A + \varepsilon, B - \varepsilon]$ is at most a constant times $h^{-2} \exp(-\varepsilon^2/2h^2)$, because $K_1 \leq A$ and $K_m \geq B$. That bound tends to zero faster than any power of h .

Multiplying by $-F$ yields the claim for $\hat{f}_{\mathbb{Q}}$ and for $\hat{f}_{\mathbb{Q}}^{\text{tw}}$. $\square$

Remark 1. *The argument treats quotes as exact. It is not a statement about observation error, and it is not a statement about the far tails outside $[A, B]$. The equivalent kernel of (16) takes negative values, so the twiced estimator is not necessarily a density.*

6.1. Numerical Illustration

The data-generating process is the stochastic-volatility model of Heston (1993) with the equity parameters of Bakshi et al. (1997): $S_0 = 100$, $r = 0.05$, $q = 0$, $T = 1/4$, $\kappa = 1.15$, $\theta = 0.04$, $\sigma = 0.39$, $\rho = -0.64$, $v_0 = 0.04$. The c.f. of $\log S_T$ uses the little-trap branch of Albrecher et al. (2007). Call prices are the damped integrand of Carr and Madan (1999) with damping $\alpha = 1.5$, evaluated by the trapezoidal rule on 4096 frequencies in $(0, 250]$; puts follow by parity. The true density is Fourier inversion of the same c.f. on the same frequency grid,

$$f_{\mathbb{Q}}(K) = \frac{1}{K\pi} \int_0^\infty \text{Re}[e^{-iu \log K} \varphi(u)] du,$$

where $\varphi(u) = \mathbb{E}^{\mathbb{Q}}[e^{iu \log S_T}]$. Two chains are used: a dense grid of $m = 256$ equally spaced strikes on $[30, 220]$ (spacing $\delta = 0.745$) and a sparse grid of $m = 32$

equally spaced strikes on $[70, 140]$ (spacing $\delta = 2.26$). Integrated squared error (ISE) is $\int_{60}^{150} (\hat{f}_Q - f_Q)^2 dK$, evaluated by the trapezoidal rule on 401 equally spaced points. Table 1 reports ISE for the three nested estimators at the oracle h^* , at the mesh rule $h = 1.2\delta$ of (17), and at the density-scale rule (18). The oracle minimizes the integral on a geometric grid in h .

Table 1: Integrated squared error on exact Heston quotes for the cubic-spline estimator. Mesh h is (17). The last pair of columns is (18). Quoted spline uses only the observed strikes. Tails adds the wings of Section 4.1. Tails+Twice applies (16).

Chain	Estimator	h^*	Oracle ISE	Mesh h	Mesh ISE	Den. h	Den. ISE
Dense	Quoted spline	0.15	9.76×10^{-10}	0.89	7.36×10^{-7}	1.88	1.33×10^{-5}
Dense	Tails	0.15	9.76×10^{-10}	0.89	7.36×10^{-7}	1.88	1.33×10^{-5}
Dense	Tails+Twice	0.39	2.68×10^{-12}	0.89	6.32×10^{-10}	1.88	1.59×10^{-7}
Sparse	Quoted spline	16.19	1.80×10^{-2}	2.71	1.04×10^{-1}	2.72	1.03×10^{-1}
Sparse	Tails	0.45	1.13×10^{-6}	2.71	5.29×10^{-5}	2.72	5.33×10^{-5}
Sparse	Tails+Twice	1.64	9.67×10^{-7}	2.71	2.66×10^{-6}	2.72	2.69×10^{-6}

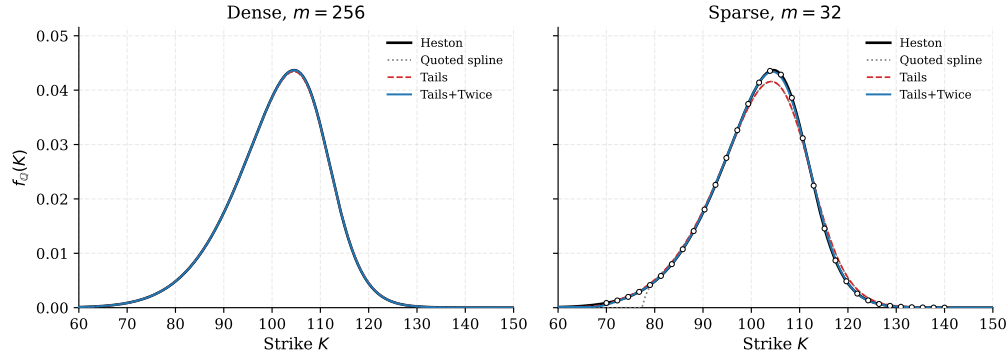


Figure 1: Heston density of S_T on $[60, 150]$. The three estimators of Table 1, each at the mesh bandwidth (17). Sparse quoted strikes are marked on the true density. Curves are truncated at zero, and the vertical scale is that of the true density.

On the dense grid the first strike lies left of the window and the last call is already zero, so the wings do not move the error. The spline without twicing has mesh integrated squared error 7.36×10^{-7} . Twicing cuts that to 6.32×10^{-10} , against an oracle 2.68×10^{-12} . The rule (18) is wider than the mesh ($h = 1.88$ against 0.89) and the twiced error is 1.59×10^{-7} . On the sparse grid the quoted spline without tails is useless: mesh integrated squared error 0.104. The left line (13) brings the

untwiced mesh error to 5.29×10^{-5} . Twicing then reaches 2.66×10^{-6} at the mesh and 2.69×10^{-6} at (18), against an oracle 9.67×10^{-7} . At this spacing the mesh bandwidth and (18) are almost the same number ($h = 2.71$ and $h = 2.72$). Figure 1 plots the three estimators at the mesh bandwidth.

Table 2 repeats the design for the variance-gamma process of Madan et al. (1998). The process parameters are those of Carr and Madan (1999): $\sigma = 0.12$, $\nu = 0.2$, $\theta = -0.14$. Spot, rate, dividend yield, and maturity are as above, as are the two strike grids and the ISE window. Variance gamma is a pure-jump Lévy process. The true density of $\log S_T$ is a modified Bessel function (Madan et al., 1998); the density of S_T is that formula times $1/K$. Calls are the discounted payoff of that density, evaluated by the trapezoidal rule.

Table 2: Integrated squared error on exact variance-gamma quotes for the cubic-spline estimator. Columns as in Table 1.

Chain	Estimator	h^*	Oracle ISE	Mesh h	Mesh ISE	Den. h	Den. ISE
Dense	Quoted spline	0.17	5.12×10^{-7}	0.89	5.08×10^{-5}	1.20	1.27×10^{-4}
Dense	Tails	0.17	5.12×10^{-7}	0.89	5.08×10^{-5}	1.20	1.27×10^{-4}
Dense	Tails+Twice	0.33	3.93×10^{-7}	0.89	8.34×10^{-6}	1.20	2.53×10^{-5}
Sparse	Quoted spline	14.03	3.27×10^{-2}	2.71	1.05×10^{-1}	1.69	1.67×10^{-1}
Sparse	Tails	0.45	3.97×10^{-5}	2.71	1.25×10^{-3}	1.69	3.63×10^{-4}
Sparse	Tails+Twice	0.80	3.31×10^{-5}	2.71	4.39×10^{-4}	1.69	1.00×10^{-4}

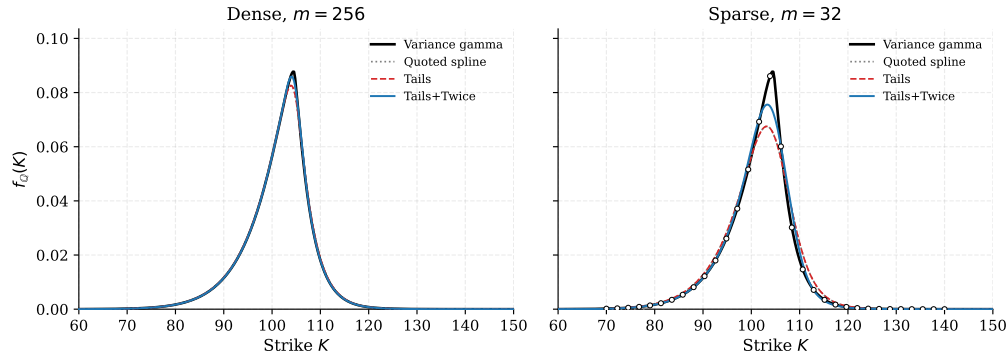


Figure 2: Variance-gamma density of S_T on $[60, 150]$. The three estimators of Table 2, each at the mesh bandwidth (17). Sparse quoted strikes are marked on the true density. Curves are truncated at zero, and the vertical scale is that of the true density.

On the dense chain the wings again do not move the digits. Twicing cuts

the mesh error from 5.08×10^{-5} to 8.34×10^{-6} , against an oracle 3.93×10^{-7} . At (18) the twiced error is 2.53×10^{-5} . The density of S_T is more peaked than under Heston, and the absolute errors are larger. On the sparse chain the quoted spline without tails has mesh error 0.105. Tails bring the untwiced mesh error to 1.25×10^{-3} , and twicing lowers it to 4.39×10^{-4} at the mesh and 1.00×10^{-4} at (18), against an oracle 3.31×10^{-5} . Here the mesh bandwidth $h = 2.71$ is wider than (18) ($h = 1.69$), so the density-scale rule is the tighter of the two. Figure 2 plots the three estimators at the mesh bandwidth.

7. Empirical Applications

7.1. Data and Methodology

The listed comparison uses European, AM-settled index options from Cboe's delayed-quotes feed (`cdn.cboe.com`). Each file is dated 23 September 2026. The S&P 500 (SPX) file is 03:56 Eastern (30,470 listed contracts across 58 expiries); the Nasdaq-100 (NDX) file is 03:44 Eastern (16,492 contracts across 46 expiries); the Russell 2000 (RUT) file is 03:44 Eastern (10,790 contracts across 30 expiries). Only the standard monthly roots SPX, NDX, and RUT are kept, not weekly SPXW, NDXW, or RUTW. Time to expiry is $T = d/365.25$ with d the calendar day count from the snapshot date. The discount rate is $r = 0.04$.

A quote is retained if the bid is positive, the ask is at least the bid, the mid is at least 0.25, and the spread is at most $\max(15, 0.002S_0, 0.5 \times \text{mid})$. The 15-point term is the tolerance on the S&P 500. On the Nasdaq-100 a fixed 15-point cap deletes puts whose spreads are a dollar or two wider and leaves a gap of several thousand points in the strike list. Among remaining put–call pairs, the forward is the median of $K + e^{rT}(C - P)$ over strikes with $|K - K_{\text{atm}}| \leq 50$ on SPX, where K_{atm} is the pair closest to the cash index; on NDX and RUT the band is 0.65% of the spot. Dividend yield is $q = r - \log(F/S_0)/T$. The working slice is out of the money (OTM): the put mid if $K \leq F$, the call mid if $K \geq F$, the missing side filled by parity. Puts with $K < 0.7F$ and mid below 0.50 are then dropped; out-of-the-money calls with mid at least 0.25 are kept. Table 3 records the filter counts. In (9) and (16), $\tilde{C} = C/(S_0 e^{-qT})$ and $F = S_0 e^{(r-q)T}$.

The December 2026 S&P 500 AM quarterly ($T = 0.235$, $S_0 = 7764.64$, $F = 7836$, $q = 0.11\%$) appears in Figure 3. After the screens it has $n = 444$ strikes on $[0.36F, 1.30F]$, of which 75% are puts. Near the money the strike mesh is 5 (0.06% of F); it is 5–25 on the put wing and up to 200 beyond $1.2F$. Median out-of-the-money mids are 39 for puts and 62 for calls. Implied volatility $\hat{\sigma}_{\text{ATM}} = 0.134$ is the median Black–Scholes implied volatility of contracts with $|K - F| \leq$

$0.03F$. The linear call tail of Section 4.1 is appended ($C_m = 0.25$ by the mid floor of the filter, $n_R = 80$ from (14)), which produces $n = 524$ for (18) and $h = 85.2$. The March 2027 AM quarterly ($T = 0.485$, $F = 7918$, $q = -0.04\%$, $\hat{\sigma}_{\text{ATM}} = 0.144$) filters to $n = 221$ strikes on $[0.20F, 1.49F]$, with a 25-point mesh at the money; (14) gives $n_R = 41$ and $h = 152.3$.

The Nasdaq-100 December 2026 AM quarterly ($T = 0.235$, $S_0 = 30732$, $F = 31025$, $q = -0.03\%$, $\hat{\sigma}_{\text{ATM}} = 0.201$) is built with the same screens. It filters to $n = 312$ strikes on $[0.29F, 1.39F]$, 83% puts; the at-the-money mesh is 25, and gaps of 1000 appear beyond $1.2F$. The bandwidth (18) is $h = 535.4$ ($n_R = 72$).

The Russell 2000 December 2026 AM quarterly ($T = 0.235$, $S_0 = 2889.92$, $F = 2912$, $q = 0.80\%$, $\hat{\sigma}_{\text{ATM}} = 0.183$) filters to $n = 99$ strikes on $[0.45F, 1.32F]$, 59% puts; the at-the-money mesh is 10. The last two calls are both 0.25, so the wing slope is the previous negative step. Equation (14) gives $n_R = 40$. The bandwidth (18) is $h = 56.2$ ($n_R = 40$).

The quoted calls are replaced by the nearest vector in Euclidean distance that is decreasing and convex in strike. Consecutive slopes lie in $[-e^{-rT}, 0]$, and each coordinate is at least the intrinsic value $e^{-rT} \max(F - K_i, 0)$. This is the unpenalized projection of Yatchew and Härdle and also the first step of Aït-Sahalia and Duarte. Puts of the projected call follow by parity, and both premiums are floored at zero.

Out-of-the-money RMSE uses the quoted mid of the put for $K \leq F$ and that of the call for $K \geq F$. Hold-out is the average of two even/odd splits (train on even strikes and score odd, then the reverse), with h recomputed from the training n and the projection recomputed from the training strikes only.

Table 3: Construction of the working slices. “At expiry” counts listed puts and calls on that root and maturity with a positive bid and ask at least the bid. “Screens” applies the mid and spread filters. “Working OTM” is the slice used by the estimators.

Slice	At expiry	Screens	Unique K	Working OTM	Puts
SPX 18 Dec 2026	927	922	473	444	335
SPX 19 Mar 2027	453	451	228	221	165
NDX 18 Dec 2026	631	631	319	312	259
RUT 18 Dec 2026	218	218	115	99	58

7.2. Estimators

Ours. Start from the projected calls of Section 7.1. Set $P_i = 1 - C_i/(S_0 e^{-qT})$. Pass a natural cubic spline through those values, with the tail knots of Section 4.1, and

invert at (18) with twicing. The listed prices are that interpolant, floored at zero, with puts by parity.

Yatchew and Härdle. Start from the projected calls of Section 7.1. A roughness penalty is then added. The objective is $\frac{1}{2}\|m - C\|^2$ plus λ times a squared second-difference penalty on the slope, scaled by the fourth power of the median gap so that λ is dimensionless. λ is chosen from $\{0\} \cup \{10^{-2}, \dots, 10^4\}$ (seven logarithmically spaced positive values) by even/odd pricing error, and the program is solved by sequential least squares. When the selected penalty is zero, the fitted call is the projection itself. Prices in Table 4 are the fitted calls at the quoted strikes, with puts by parity, floored at zero. The mass column is e^{rT} times the second difference of the fitted call, integrated as $\int \max(\hat{q}, 0)$ on the window used for every row.

Ait-Sahalia and Duarte. Start from the projected calls of Section 7.1. The call used for pricing is a local linear regression of those values, with Gaussian weights. Its bandwidth is chosen by even/odd pricing error from the grid 0.25, 0.5, 1, 2, 4 times $F\hat{\sigma}\sqrt{T}n^{-1/5}$. The density drawn in Figure 3 is e^{rT} times twice the quadratic coefficient of a local cubic of the same list, at the fixed second-derivative bandwidth $0.9F\hat{\sigma}\sqrt{T}n^{-1/9}$. Prices in the table are the local-linear calls, floored, with puts by parity.

Bondarenko PCA. Start from the projected calls of Section 7.1. The density is a mixture of 41 lognormals with a common log-width h and nonnegative weights. The centers are equally spaced in log strike from the first quoted strike to the last. Each component prices a call in closed form, Black–Scholes with forward $e^{z_j+h^2/2}$. The weights minimise the sum of squared errors in those projected calls, plus two penalties, one for $\sum_j a_j = 1$ and one for $\sum_j a_j e^{z_j+h^2/2} = F$. Nonnegative least squares returns the weights. They are not divided by their sum afterwards; a weight above one is clipped to one. The width h is chosen by even/odd pricing error on a grid of multiples of the center spacing. Prices in Table 4 are those closed-form calls, floored, with puts by parity.

Priestley and Chao. Start from the projected calls of Section 7.1. Section 3.2 records why (11) is a poor estimator of a call: the level C_i is large deep in the money, where C'' is nearly zero, and the spacing weight does not remove that level. The listed estimator is the cubic that section substitutes, (12): a clamped cubic spline through the projected calls, extended by linear wings of slope in $[-e^{-rT}, 0]$, then convolved with a Gaussian kernel. The bandwidth is (18), the same number our estimator uses on that fit. A bandwidth chosen by even/odd

pricing error on these chains is a fraction of the median strike gap, and the second derivative at that width is not a smooth density. Prices in the table are the smoothed call at the quoted strikes, floored, with puts by parity. The density in Figure 3 is (12) at (18).

Hold-out refits the projection and every tuned method on the training strikes only, including the choice of its tuning constant, and scores the other half against the quoted mids. The two directions are averaged.

The solved Yatchew and Härdle projection is the nearest decreasing convex call. Its out-of-the-money error is 0.022 on SPX December, 0.024 on SPX March, 0.785 on NDX December, and 0.026 on RUT December. On SPX December the cross-validated penalty is zero, so 0.022 is the distance from the quotes to the cone. Any smoother that stays inside the cone, including Aït-Sahalia and Duarte, lies at or above it. Their cross-validated local linear is 0.337, 0.596, 2.499, and 0.339. With centers on the quoted log-strike range and a width chosen by pricing error, PCA out-of-the-money error is 0.038 on SPX December, 0.079 on SPX March, 1.147 on NDX December, and 0.034 on RUT December. The Priestley and Chao smoother has out-of-the-money error 1.721, 3.022, 12.270, and 1.630. The estimator of this paper is the Gaussian inversion of the cubic spline of P , with tails and twicing, at (18). The pricing error is 0.051, 0.105, 1.073, and 0.050. The hold-out error is 0.080, 0.165, 1.177, and 0.072.

Averaging the out-of-the-money hold-out error across the four chains, one obtains 0.374 for the estimator of this paper, 0.409 for Yatchew and Härdle, 1.245 for Aït-Sahalia and Duarte, 0.336 for PCA, and 6.130 for Priestley and Chao. Weighting the average with the inverse forward, normalized to sum to one, one has 0.149, 0.189, 0.671, 0.105, and 3.220, in the same order.

Table 4: Out-of-the-money repricing. Every fit starts from the projection in Section 7.1 and is scored on the quoted mids. Ours is the cubic-spline inversion at (18) ($h = 85.2, 152.3, 535.4, 56.2$). Yatchew and Härdle is that projection plus its cross-validated penalty. Ait-Sahalia and Duarte is priced from the local linear at its cross-validated bandwidth. PCA is priced from the lognormal mixture. Priestley and Chao is the cubic (12) at the same bandwidth. OTM is put RMSE for $K \leq F$ and call RMSE for $K \geq F$, premiums floored at zero. Mass is $\int \max(\hat{q}, 0)$ on $[\max(50, 0.2F), 2.4F]$. Hold-out refits the projection and the tuning on each half. Average is the equal-weight mean of the four chains. Weighed av. weights chain i by $1/F_i$, normalized to sum to one.

Chain	Method	RMSE OTM	Puts	Calls	Hold-out	Mass
SPX Dec	Ours	0.051	0.041	0.073	0.080	1.00
SPX Dec	Yatchew and Härdle	0.022	0.022	0.020	0.050	0.97
SPX Dec	Ait-Sahalia and Duarte	0.337	0.252	0.518	0.444	1.01
SPX Dec	PCA	0.038	0.035	0.046	0.040	1.00
SPX Dec	Priestley and Chao	1.721	1.268	2.669	2.379	1.00
SPX Mar	Ours	0.105	0.078	0.161	0.165	1.00
SPX Mar	Yatchew and Härdle	0.024	0.022	0.031	0.145	1.00
SPX Mar	Ait-Sahalia and Duarte	0.596	0.401	0.963	0.789	1.01
SPX Mar	PCA	0.079	0.063	0.115	0.079	1.00
SPX Mar	Priestley and Chao	3.022	2.001	4.925	3.986	1.00
NDX Dec	Ours	1.073	1.137	0.672	1.177	1.00
NDX Dec	Yatchew and Härdle	0.785	0.839	0.432	1.285	1.00
NDX Dec	Ait-Sahalia and Duarte	2.499	2.418	2.858	3.278	1.02
NDX Dec	PCA	1.147	1.131	1.221	1.187	1.00
NDX Dec	Priestley and Chao	12.270	10.953	17.324	16.113	1.00
RUT Dec	Ours	0.050	0.045	0.055	0.072	1.00
RUT Dec	Yatchew and Härdle	0.026	0.022	0.031	0.155	1.00
RUT Dec	Ait-Sahalia and Duarte	0.339	0.293	0.394	0.467	1.09
RUT Dec	PCA	0.034	0.030	0.039	0.038	1.00
RUT Dec	Priestley and Chao	1.630	1.394	1.914	2.041	1.00
Average	Ours	0.320	0.326	0.240	0.374	1.00
Average	Yatchew and Härdle	0.214	0.226	0.129	0.409	0.99
Average	Ait-Sahalia and Duarte	0.943	0.841	1.183	1.245	1.03
Average	PCA	0.325	0.315	0.355	0.336	1.00
Average	Priestley and Chao	4.661	3.904	6.708	6.130	1.00
Weighed av.	Ours	0.113	0.107	0.112	0.149	1.00
Weighed av.	Yatchew and Härdle	0.064	0.064	0.049	0.189	0.99
Weighed av.	Ait-Sahalia and Duarte	0.501	0.415	0.660	0.671	1.05
Weighed av.	PCA	0.101	0.094	0.116	0.105	1.00
Weighed av.	Priestley and Chao	2.472	1.980	3.460	3.220	1.00

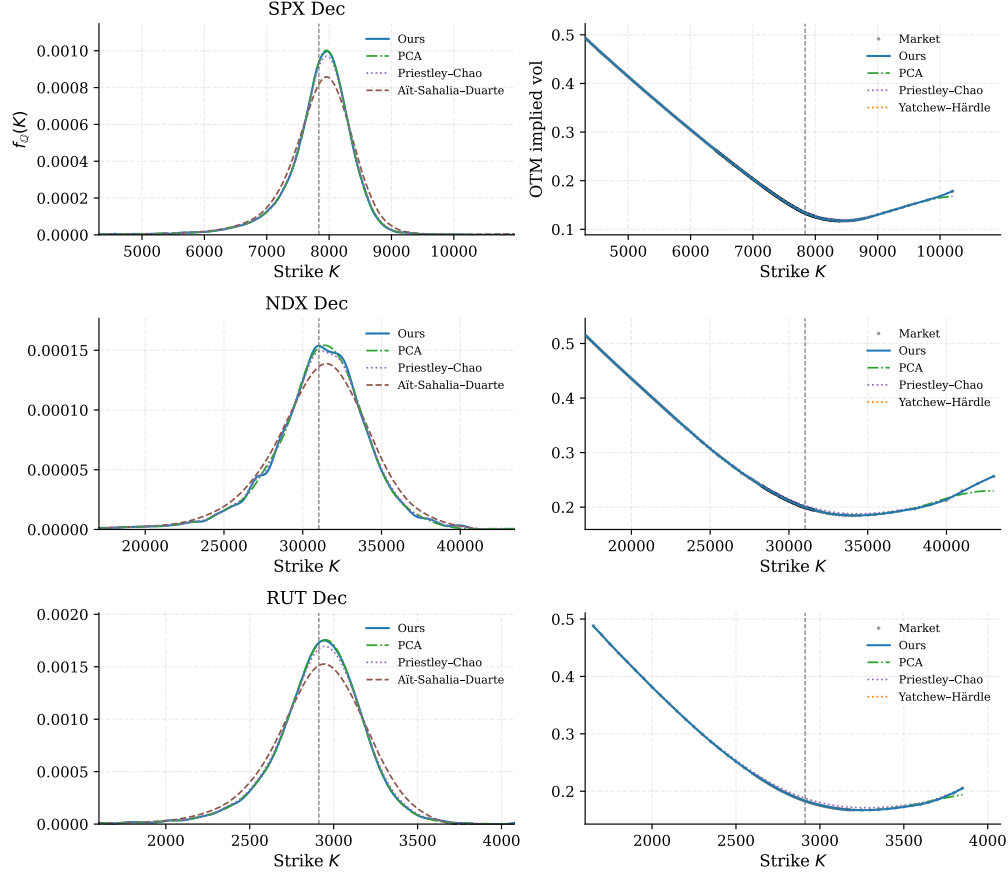


Figure 3: SPX, NDX, and RUT, 18 December 2026, snapshot 23 September 2026. Each curve is fit to the projected call of Section 7.1. Left: $\max(\hat{f}_Q, 0)$. The cubic-spline estimator is at (18). Ait-Sahalia and Duarte is the local cubic at $0.9 F \hat{\sigma} \sqrt{T} n^{-1/9}$. PCA is the fitted lognormal mixture. Priestley and Chao is (12) at (18). Right: out-of-the-money implied volatility from the same fits' prices, with Yatchew and Härdle in place of the local cubic. The dashed line is the forward.

8. Conclusion

The construction in the paper produces both an in-fill consistent estimate of the RND and a pricing interpolant. In terms of repricing accuracy, it improves upon a naive kernel smoother by a large margin and is competitive with other established methods. The Yatchew and Härdle projection remains slightly closer in sample (but cannot produce a density), and PCA is marginally closer on the weighted hold-out (but is not consistent without modifications).

Code and Data Availability

A Python implementation of (16) and a replication script are available at <https://github.com/s-broda/RND>.

References

- Aït-Sahalia, Y., Duarte, J., 2003. Nonparametric option pricing under shape restrictions. *Journal of Econometrics* 116, 9–47.
- Aït-Sahalia, Y., Lo, A.W., 1998. Nonparametric estimation of state-price densities implicit in financial asset prices. *Journal of Finance* 53, 499–547.
- Aït-Sahalia, Y., Lo, A.W., 2000. Nonparametric risk management and implied risk aversion. *Journal of Econometrics* 94, 9–51.
- Albrecher, H., Mayer, P., Schoutens, W., Tistaert, J., 2007. The little Heston trap. *Wilmott Magazine* , 83–92.
- Anagnou, I., Bedendo, M., Hodges, S., Tompkins, R., 2005. Forecasting accuracy of implied and GARCH-based probability density functions. *Review of Futures Markets* 11, 41–66.
- Bahra, B., 1997. Implied Risk-Neutral Probability Density Functions from Option Prices: Theory and Application. Technical Report Working Paper 66. Bank of England.
- Bakshi, G., Cao, C., Chen, Z., 1997. Empirical performance of alternative option pricing models. *Journal of Finance* 52, 2003–2049.
- Birke, M., Pilz, K.F., 2009. Nonparametric option pricing with no-arbitrage constraints. *Journal of Financial Econometrics* 7, 53–76.
- Bliss, R.R., Panigirtzoglou, N., 2002. Testing the stability of implied probability density functions. *Journal of Banking & Finance* 26, 381–422.
- Bondarenko, O., 2003. Estimation of risk-neutral densities using positive convolution approximation. *Journal of Econometrics* 116, 85–112.
- Breeden, D.T., Litzenberger, R.H., 1978. Prices of state-contingent claims implicit in option prices. *Journal of Business* 51, 621–651.

- Buchen, P.W., Kelly, M., 1996. The maximum entropy distribution of an asset inferred from option prices. *Journal of Financial and Quantitative Analysis* 31, 143–159.
- Carr, P., Madan, D., 1999. Option valuation using the fast Fourier transform. *Journal of Computational Finance* 2, 61–73.
- Cont, R., Tankov, P., 2004. Non-parametric calibration of jump-diffusion option pricing models. *Journal of Computational Finance* 7, 1–49.
- Cooper, N., 1999. Testing techniques for estimating implied RNDs from the prices of European-style options, in: *Proceedings of the BIS Workshop on Estimating and Interpreting Probability Density Functions*. Bank for International Settlements.
- Corrado, C.J., Su, T., 1996. Skewness and kurtosis in S&P 500 index returns implied by option prices. *Journal of Financial Research* 19, 175–192.
- Coutant, S., Jondeau, E., Rockinger, M., 2001. Reading PIBOR futures options smiles: The 1997 snap election. *Journal of Banking & Finance* 25, 1957–1987.
- Dupire, B., 1994. Pricing with a smile. *Risk* 7, 18–20.
- Fan, J., Mancini, L., 2009. Option pricing with model-guided nonparametric methods. *Journal of the American Statistical Association* 104, 1351–1372.
- Fengler, M.R., 2005. *Semiparametric Modeling of Implied Volatility*. Springer, Berlin.
- Fengler, M.R., 2009. Arbitrage-free smoothing of the implied volatility surface. *Quantitative Finance* 9, 417–428.
- Figlewski, S., 2010. Estimating the implied risk neutral density for the U.S. market portfolio, in: Bollerslev, T., Russell, J., Watson, M. (Eds.), *Volatility and Time Series Econometrics: Essays in Honor of Robert Engle*. Oxford University Press, pp. 323–358.
- Gatheral, J., Jacquier, A., 2014. Arbitrage-free SVI volatility surfaces. *Quantitative Finance* 14, 59–71.

- Grith, M., Härdle, W., Schienle, M., 2012. Nonparametric estimation of risk-neutral densities, in: Duan, J.C., Härdle, W.K., Gentle, J.E. (Eds.), *Handbook of Computational Finance*. Springer, pp. 277–305.
- Heston, S.L., 1993. A closed-form solution for options with stochastic volatility with applications to bond and currency options. *Review of Financial Studies* 6, 327–343.
- Jackwerth, J.C., Rubinstein, M., 1996. Recovering probability distributions from option prices. *Journal of Finance* 51, 1611–1631.
- Jarrow, R., Rudd, A., 1982. Approximate option valuation for arbitrary stochastic processes. *Journal of Financial Economics* 10, 347–369.
- Jondeau, E., Rockinger, M., 2000. Reading the smile: The message conveyed by methods which infer risk neutral densities. *Journal of International Money and Finance* 19, 885–915.
- Lai, W.N., 2014. Comparison of methods to estimate option implied risk-neutral densities. *Quantitative Finance* 14, 1839–1855.
- Lewis, A.L., 2000. *Option Valuation under Stochastic Volatility*. Finance Press, Newport Beach.
- Liu, X., Shackleton, M.B., Taylor, S.J., Xu, X., 2007. Closed-form transformations from risk-neutral to real-world distributions. *Journal of Banking & Finance* 31, 1501–1520.
- Lu, S., 2019. Monte Carlo analysis of methods for extracting risk-neutral densities with affine jump diffusions. *Journal of Futures Markets* 39, 1587–1612.
- Ludwig, M., 2015. Robust estimation of shape-constrained state price density surfaces. *Journal of Derivatives* 22, 56–72.
- Madan, D.B., Carr, P.P., Chang, E.C., 1998. The variance gamma process and option pricing. *European Finance Review* 2, 79–105.
- Malz, A.M., 1997. Estimating the probability distribution of the future exchange rate from option prices. *Journal of Derivatives* 5, 18–36.

- Melick, W.R., Thomas, C.P., 1997. Recovering an asset's implied PDF from option prices: An application to crude oil during the Gulf crisis. *Journal of Financial and Quantitative Analysis* 32, 91–115.
- Monnier, J.B., 2014. Risk-neutral density recovery via spectral analysis. *SIAM Journal on Financial Mathematics* 5, 558–587. doi:10.1137/110840340.
- Priestley, M.B., Chao, M.T., 1972. Non-parametric function fitting. *Journal of the Royal Statistical Society, Series B* 34, 385–392.
- Ritchey, R.J., 1990. Call option valuation for discrete normal mixtures. *Journal of Financial Research* 13, 285–296.
- Rompolis, L.S., Tzavalis, E., 2007. Retrieving risk neutral densities based on risk neutral moments through a Gram–Charlier series expansion. *Mathematical and Computer Modelling* 46, 225–234.
- Ross, S.A., 1976. Options and efficiency. *Quarterly Journal of Economics* 90, 75–89.
- Rubinstein, M., 1994. Implied binomial trees. *Journal of Finance* 49, 771–818.
- Santos, A.A.P., Guerra, J., 2015. Implied risk neutral densities from option prices: Hypergeometric, spline, lognormal, and Edgeworth functions. *Journal of Futures Markets* 35, 655–678.
- Schucany, W.R., Sommers, J.P., 1977. Improvement of kernel type density estimators. *Journal of the American Statistical Association* 72, 420–423.
- Shimko, D., 1993. Bounds of probability. *Risk* 6, 33–37.
- Silverman, B.W., 1986. *Density Estimation for Statistics and Data Analysis*. Chapman & Hall, London.
- Söderlind, P., Svensson, L.E.O., 1997. New techniques to extract market expectations from financial instruments. *Journal of Monetary Economics* 40, 383–429.
- Stutzer, M., 1996. A simple nonparametric approach to derivative security valuation. *Journal of Finance* 51, 1633–1652.
- Xiu, D., 2014. Hermite polynomial based expansion of European option prices. *Journal of Econometrics* 179, 158–177.

Yatchew, A., Härdle, W., 2006. Nonparametric state price density estimation using constrained least squares and the bootstrap. *Journal of Econometrics* 133, 579–609.